

The Evolution of Artificial Intelligence

From Dreams to Reality

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Today's Roadmap

- 1 The AI Effect
- 2 Why Booms and Winters?
- 3 1950–1956: The Beginning
- 4 Symbolic AI Era (1956–1980)
- 5 First AI Winter (1974–1980)
- 6 Expert Systems (1980s)
- 7 Second AI Winter (1987–1993)
- 8 Statistical AI Era (1990s–2010)
- 9 Deep Learning & Transformers
- 10 Foundation Models & Generative AI (2022–Present)
- 11 Summary

“AI is whatever hasn’t been solved yet.”

- Whenever a machine solves a hard problem, people stop calling it AI.
- Instead, it becomes “just an algorithm” or “just software.”

The AI Effect: Examples

Once considered AI:

- Playing chess
- Optical Character Recognition (OCR)
- Speech recognition
- Route planning
- Spam filtering
- Recommendation systems

Now considered:

- A search algorithm
- A software feature
- An everyday utility

Definition

This phenomenon is called the **AI Effect** – once solved, a problem is no longer perceived as AI.

Quick Poll: Is It AI?

For each of these, ask yourselves: *is this “AI” today, or has it already become “just software”?*

- Google Maps route planning
- Autocorrect on your phone
- Face unlock
- ChatGPT / Claude answering questions
- Self-driving cars
- Netflix recommendations

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Notice: some feel “ordinary,” others still feel “magical.” That feeling is exactly the AI Effect in motion.

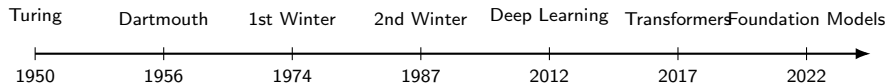
The Cycle of Hype and Disappointment

- Researchers are ambitious: build machines with human-level intelligence.
- Ambition → optimistic predictions.
- Optimism → funding, investment, public attention.
- Reality falls short → enthusiasm declines → funding cut.

Key Idea

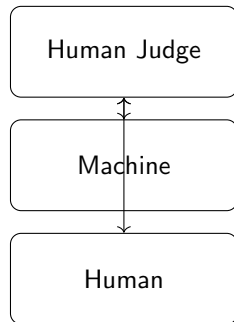
The history of AI is a sequence of **booms and winters**, driven by the gap between *expectation* and *reality*.

Timeline Preview



1950 – Alan Turing

- Alan Turing asked: *“Can machines think?”*
- Proposed the **Turing Test**.
- If a machine's conversation is indistinguishable from a human's, it can be considered intelligent.



1956 – Birth of “Artificial Intelligence”

- **Dartmouth Summer Research Project** on Artificial Intelligence.
- **John McCarthy** coins the term *Artificial Intelligence*.
- Belief: human-level intelligence achievable within a few decades.

This marked the birth of AI as an academic discipline.

Symbolic AI: The Core Idea

Central Idea

Represent knowledge as **symbols** and manipulate those symbols using **logical rules**.

Major components:

- Logic-based reasoning
- Predicate (First-Order) Logic
- Knowledge Representation
- Rule-based Systems
- Search Algorithms
- Planning

Symbolic AI: A Worked Example

Goal: derive new facts from known facts using logical rules.

Given

- All humans are mortal. $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$
- Socrates is human. $\text{Human}(\text{Socrates})$

Derived by the system (not told directly!)

Socrates is mortal. $\text{Mortal}(\text{Socrates})$

This is the essence of Symbolic AI: new knowledge emerges purely from symbol manipulation, with no data or learning involved.

Symbolic AI: Application Areas

- Machine Translation
- Game Playing
- Automated Theorem Proving
- Robotics

The future looked very promising...

1966 – Machine Translation Disappointment

- Grand challenge: automatic translation between languages.
- Technology was far from ready.
- **ALPAC Report (1966)**: concluded MT had not achieved expected progress.
- Result: significant funding cuts.

1969 – Neural Networks Face a Setback

- Marvin Minsky & Seymour Papert analyze the **single-layer perceptron**.
- Showed it cannot solve non-linearly separable problems (e.g., **XOR problem**).
- Neural network research loses momentum for years.

Important Note

This was **not** the cause of the First AI Winter – but it did reduce interest in neural networks.

First AI Winter: Causes

- Machine Translation failed to meet expectations.
- Symbolic AI struggled with complex real-world problems.
- Computers lacked sufficient processing power & memory.
- Researchers overestimated the pace of progress.

Governments and industries reduced funding $\Rightarrow$ **First AI Winter**.

The Rise of Expert Systems

- AI's remarkable comeback in the early 1980s.
- Capture human expert knowledge as thousands of **IF-THEN** rules.

Example Rules

- IF fever AND cough → Flu
- IF battery voltage is low → Replace battery

Application domains: Medical diagnosis, Financial decisions, Industrial troubleshooting, Computer configuration.

Expert Systems: The Ecosystem

- Rapid industry growth; heavy investment.
- **Lisp** became the dominant AI programming language.
- Specialized **Lisp Machines** were built to run AI applications.

Why Expert Systems Failed to Scale

- Thousands of rules had to be written manually.
- Maintaining large rule bases was extremely expensive.
- Systems could not learn from new data.
- Performed well only in narrow domains.
- Failed to generalize beyond expert-provided knowledge.

Knowledge Acquisition Bottleneck

The core problem: knowledge had to be manually extracted and encoded from human experts.

Second AI Winter: Aftermath

- Cheap Unix workstations replace expensive Lisp Machines.
- Investment declines sharply as expectations go unmet.
- The term “Artificial Intelligence” becomes **unpopular** in industry.

Rebranded research areas:

- Machine Learning • Pattern Recognition
- Statistical NLP • Data Mining

Why Did the Paradigm Shift Happen?

Two winters taught the field a hard lesson: hand-written rules do not scale.

What changed by the 1990s:

- Digitization → text, speech, image **data** became available.
- Cheaper computing (workstations, later PCs).
- Growing use of the internet.

The new bet:

- Stop hand-coding knowledge.
- Let algorithms **infer patterns** from examples.
- Probability & statistics replace hard logical rules.

Cause and Effect

Data availability + cheaper compute $\Rightarrow$ learning from data became more practical than encoding expert knowledge by hand.

From Knowledge Engineering to Learning from Data

- Shift: writing rules manually $\rightarrow$ learning patterns from data.

Major developments:

- Bayesian Networks
- Hidden Markov Models
- Support Vector Machines
- Decision Trees
- Reinforcement Learning
- Markov Decision Processes

2012 – The Deep Learning Revolution

- Deep neural networks show breakthrough performance in computer vision.
- Landmark moment: a deep convolutional network (AlexNet) dramatically outperforms all prior methods on the ImageNet image-recognition benchmark.

Enabling factors:

- Large labeled datasets (e.g., millions of images in ImageNet)
- Powerful GPUs – parallel hardware originally built for graphics
- Better optimization algorithms

Notice the pattern: same idea (neural networks) as the 1960s/80s – but now with the data and compute to actually make it work.

Bridging Idea: History Repeats

Recall Minsky & Papert (1969) – neural networks were dismissed for decades.

1969

- Tiny datasets
- No GPUs
- Shallow networks only

2012

- Millions of images
- GPU acceleration
- Deep, many-layer networks

Lesson

An idea being “wrong” in one era can simply mean it was **ahead of its infrastructure.**

2017 – The Transformer Architecture

- Introduced in 2017; core innovation is the **attention mechanism**.
- Attention lets the model weigh the relevance of every word to every other word, regardless of distance in the sentence.
- This solved a key weakness of earlier sequential models (RNNs), which struggled with long-range dependencies.
- Became the foundation for modern NLP.

Nearly every state-of-the-art language model today is Transformer-based.

The Age of Foundation Models

- Large Language Models: GPT, LLaMA, Gemini, Claude, Qwen, ...

Capabilities:

- Understand natural language
- Generate text & write code
- Summarize documents
- Answer questions
- Translate languages
- Reason over complex tasks

For the first time: general-purpose AI assistants using a single model.

Evolution of AI: Timeline of Paradigms

Period	Dominant Paradigm
1956–1985	Symbolic AI (Logic, Search, KR, Expert Systems)
1990–2010	Statistical AI (Bayesian Nets, HMMs, SVMs, RL)
2012–Present	Deep Learning
2017–Present	Transformers
2022–Present	Foundation Models & Generative AI

Discussion: Where Are We Headed?

Recall our opening line: *“AI is whatever hasn’t been solved yet.”*

Think, pair, share (2 minutes):

- Which of today’s “AI” capabilities (reasoning, coding assistants, autonomous agents) do you think will feel like “ordinary software” in 10 years?
- Are we in another hype phase right now – heading toward a **Third AI Winter**, or is this time genuinely different? Why?

Artificial Intelligence is not a fixed destination – it is a moving frontier.

- Every generation of researchers pushes the boundary of what machines can do.
- The **AI Effect** we opened with is the same force throughout this entire history: chess, OCR, and expert systems were all once “true AI” – and were each demoted to “ordinary software” in turn.
- Today’s frontier – reasoning, autonomous agents, multimodal systems – may one day suffer the very same fate.

Thank You!

Questions & Discussion